1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

CEACAM5 overexpression in MSI-H bladder tumors is associated with high Tumor-Infiltrating Lymphocyte (TIL) density but impairs cytotoxic T cell effector function via CEACAM5-mediated disruption of the immune synapse, producing a functionally exhausted infiltrate. Consequently, MSI-H bladder tumors with high CEACAM5 expression and high TIL density will exhibit lower therapy_response (to immune checkpoint blockade) than MSI-H tumors with low CEACAM5 expression, despite comparable TIL levels.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CEACAM5, MSI-H
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CEACAM5: CEACAM5 is upregulated in bladder tumors that show high immune cell infiltration and co-occur with MSI-H, but high immune infiltration in MSI-H tumors does not always translate into effective immunotherapy benefit—why might an MSI-H, TIL-rich tumor fail to respond? 2) Tier 1 path used: the single path linking CEACAM5 upregulation → higher immune cell infiltration → MSI-H; I build on CEACAM5 as the putative tumor-intrinsic modulator and MSI-H as the genomic context that drives TIL recruitment.

MSI-H: MSI-H, but high immune infiltration in MSI-H tumors does not always translate into effective immunotherapy benefit—why might an MSI-H, TIL-rich tumor fail to respond? 2) Tier 1 path used: the single path linking CEACAM5 upregulation → higher immune cell infiltration → MSI-H; I build on CEACAM5 as the putative tumor-intrinsic modulator and MSI-H as the genomic context that drives TIL recruitment.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Higher immune cell infiltration	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: in-silico analyses in TCGA-BLCA to assess interaction between CEACAM5 expression, MSI-H status, TIL density, and therapy_response proxies;

Prediction 2: mechanistic validation in vitro using bladder cancer cell lines with CEACAM5 knockdown/overexpression co-cultured with cytotoxic T cells to measure killing, degranulation, and cytokine production;

Prediction 3: ex vivo/in situ validation with multiplex immunofluorescence to correlate CEACAM5, TIL phenotype, and synapse markers;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Higher immune cell infiltration: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.